SMCNet: Sparse-Inspired Masked Convolutional Network for Hyperspectral Anomaly Detection

Zhaoyue Wu, *Graduate Student Member, IEEE*, Mercedes E. Paoletti, *Senior Member, IEEE*, Hongjun Su, *Senior Member, IEEE*, Pan Zheng, *Graduate Student Member, IEEE*, Zhong Zhang, Juan M. Haut, *Senior Member, IEEE*, and Antonio Plaza, *Fellow, IEEE*

***Abstract*— Hyperspectral anomaly detection, which aims to search and localize potential targets, is a research area with extensive application prospects and profound implications. In recent years, the emergence of unsupervised and self-supervised deep learning for image reconstruction has provided inspiring solutions for hyperspectral anomaly detection. However, due to sensor-induced and environmental effects, the full-image detection networks inevitably reconstruct anomalies along with the background. Existing detectors indirectly mitigate anomaly reconstruction by imposing constraints on hidden features or loss functions, but they provide unsatisfactory performance in large target detection scenarios. This work straightforwardly addresses this issue from the input source, i.e., introducing the concept of masked autoencoders (MAEs) into fully convolutional networks and further developing a sparse-inspired masked convolutional network (SMCNet) consisting of three mutually supportive components: 1) a hierarchical encoder; 2) a sparse projection layer; and 3) a hierarchical decoder. The encoder employs an adaptive potential anomaly masking strategy, leveraging sparse convolution for extracting multidimensional features of the remaining background. Meanwhile, a sparse-guided projection layer is created by discarding the positional embedding technique to populate the uncoded region and guide the background recovery without introducing anomalies. Finally, the decoder couples the hierarchical structure and a hybrid attention mechanism (local–middle–global and spatial–spectral) to refine the background during image recovery, whereas anomalies in the residual map are highlighted. Extensive experiments using ten typical competitors on six different types of datasets validate the effectiveness and generalization ability of the newly proposed SMCNet method.**

Received 4 June 2024; revised 7 August 2024 and 17 September 2024; accepted 7 October 2024. Date of publication 9 October 2024; date of current version 21 October 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 42122008 and Grant 42371327; in part by the Consejería de Economía, Ciencia y Agenda Digital of the Junta de Extremadura and the European Regional Development Fund (ERDF) of the European Union, under Grant GR21040; in part by the ERDF of the European Union Interreg V-A Espana-Portugal (POCTEP) 2021–2027 Program under Grant 0206_RAT_EOS_PC_6_E; and in part by China Scholarship Council under Grant 202106710022. *(Corresponding author: Hongjun Su.)*

Zhaoyue Wu, Mercedes E. Paoletti, Juan M. Haut, and Antonio Plaza are with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, University of Extremadura, 10003 Caceres, Spain (e-mail: zhaoyue_wu@163.com; mpaoletti@unex.es; juanmariohaut@unex.es; aplaza@unex.es).

Hongjun Su is with the College of Geography and Remote Sensing, Hohai University, Nanjing 211100, China (e-mail: hjsu@hhu.edu.cn).

Pan Zheng is with the School of Earth Sciences and Engineering, Hohai University, Nanjing 211100, China (e-mail: zhengpan1013@163.com).

Zhong Zhang is with the School of Public Administration, China University of Geosciences, Wuhan 430074, China (e-mail: zhangzhong@cug.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3477256

***Index Terms*— Deep learning, hybrid attention mechanism, hyperspectral anomaly detection, masked convolutional network, sparse representation.**

I. Introduction

HYPERSPECTRAL imaging, as a technology capable of providing continuous spectral and fine-grained spatial information about ground objects, has become one of the most significant data sources for contemporary Earth observation [1], [2], [3]. Hyperspectral anomaly detection, which focuses on identifying regions within an image that exhibit different physical and chemical properties compared to the dominant features, is a fundamental and promising research topic [4], [5]. With the absence of prior knowledge, the main goal of hyperspectral anomaly detection is to explore and pinpoint potential targets while removing other extraneous backgrounds [6], [7]. This technique plays a crucial role in border security, resource surveying, precision agriculture, disaster relief, environmental surveillance, and other applications [8].

In this context, hyperspectral anomaly detection is basically a form of "no reference detection," confronting various challenges and complexities [9], [10]. Primarily, the processing and analysis of high-dimensional data require powerful computational capabilities [11]. Moreover, spectral mixing, noise, and changes in environmental conditions increase the difficulty of identifying anomalous pixels. Therefore, designing efficient approaches for addressing these problems has become a challenge [12], [13]. Briefly, hyperspectral anomaly detection methods can be categorized into traditional and deep learning methods. Further, from the perspective of problem-solving ideas, they can be subdivided into three categories: statistics-based, feature mining-based, and image reconstruction-based.

A. Insight Into Traditional Hyperspectral Anomaly Detection

Traditional methods for hyperspectral anomaly detection were dominant until 2018, giving rise to a plethora of algorithms based on statistical theory, sparse representation, and tensor decomposition. The Reed–Xiaoli (RX) algorithm is a milestone work that benefited from statistical theory, assuming that background pixels follow a simple multivariate Gaussian distribution while anomalous pixels deviate from it [14]. Several examples have demonstrated that RX can achieve

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

desirable results with simple backgrounds rather than with complex ones [5], [24]. In general, RX performs unsatisfactorily for anomaly detection in complex urban areas and better in natural scenes. Subsequently, corresponding improvements have been developed, such as localized RX (LRX) [15], Gaussian mixture modeling [16], and nonlinear mapping [17], [18]. With the development of compressive sensing theories, the collaborative representation derived from sparse representation theory was applied to hyperspectral anomaly detection, opening up a new research avenue and attracting considerable attention. In the collaborative representation detector (CRD), each pixel under test (PUT) is linearly reconstructed (approximately) from the surrounding background pixels, and the reconstruction residual is the anomaly degree of the PUT [19]. In this case, the surrounding background pixels are sampled by a dual window. Under ideal conditions, where all samples belong to the background, anomalous targets can be easily detected. However, in practical applications, anomalous targets are complex and diverse, making it difficult to obtain pure background samples through a simple dual-window strategy. Hence, statistical distributions [20], principal components analysis (PCA) [21], and density clustering are combined with the dual window [22] to mitigate anomalous contamination. Moreover, a recent work comprehensively improved the CRD by adopting a relaxed implementation from both model optimization and dictionary construction perspectives, significantly enhancing anomaly detection performance [23].

Unlike the aforementioned methods, anomaly detectors based on tensor decomposition split the original image into a low-rank background tensor and a sparse anomaly tensor, taking into account the inherent spatial and spectral structures of hyperspectral images. Depending on the formation mode of the background tensor or the anomaly tensor, two methods can be applied, i.e., direct and indirect decomposition. Direct decomposition imposes regularization constraints directly on the background and anomaly tensor. Examples include the robust PCA (RPCA), tensor RPCA [24], and prior-based tensor approximation (PTA) [25], [26], [27], [28]. In contrast, indirect decomposition performs image reconstruction through a background or anomaly dictionary and imposes regularization constraints on the corresponding representation coefficient tensor. Examples include the low-rank and sparse representation [29], graph and total variation regularized low-rank representation [30], generalized shrinkage mappings for hyperspectral anomaly detection [31], and tensor low-rank and sparse representation [32]. Besides, leveraging the powerful learning capacity of deep learning to boost the solution provided by the alternating direction method of multipliers has emerged as a new trend [33], [34]. Conclusively, most of the traditional hyperspectral anomaly detection methods are based on statistics and image reconstruction strategies, where the breakthrough difficulty lies in how to design a reasonable mathematical model as well as to construct a suitable dictionary. Further, these methods only utilize shallow features to perform downstream tasks and have not yet explored deep features, which are crucial for anomaly detection in complex scenes.

B. Deep Learning and Hyperspectral Processing: Exploring New Horizons in Anomaly Detection

Since the breakthrough achievement of AlexNet in 2012, a deep convolutional neural network (CNN), the research fervor surrounding deep learning has extended to a variety of applications. Initial CNN-based anomaly detection methods were implemented by transfer learning due to the lack of training samples [35], which severely limited the scalability of existing methods. Until recent years, deep learning methods have been widely applied to hyperspectral anomaly detection tasks due to the progressive development of unsupervised and self-supervised deep learning techniques, including autoencoders (AEs) and generative adversarial networks (GANs). Regarding the former, AE-based hyperspectral anomaly detection approaches endeavor to compress the input image into a low-dimensional latent space through an encoder and then reconstruct the background from the latent space using a decoder [8]. Ideally, the latent space is formed by background vectors, in which case the decoder reconstructs the background satisfactorily and the reconstructed residuals are used to determine anomalies. To this end, several strategies have been explored. For instance, an adaptive-weight loss function has been designed to weaken the reconstruction of anomalies in an autonomous hyperspectral anomaly detection (Auto-AD) network based on fully convolutional AEs [36]. Graph regularization has also been imposed on the latent space to mitigate anomalous contamination [37]. No reference-spot and no reference-block sampling strategies have been devised to build pure background samples [38], [39], [40]. A background-guided pattern has also been developed to facilitate background reconstruction [41]. Two decoders have been adopted to separate low-rank background and sparse anomalies [42].

On the other hand, GAN-based hyperspectral anomaly detection methods train the generator to produce fake samples that adhere to the background distribution and gradually complete the image reconstruction process until the discriminator cannot distinguish between fake-synthetic and true samples. How to motivate the generator to produce background and ignore anomalies is critical to the success of such methods. Until now, background distribution assumptions [43], background sample searching [44], [45], the initial separation between background and anomalies [46], [47], and regularization constraints [48] have been extensively studied to minimize anomaly reconstruction to some extent. Moreover, the Transformer has been introduced into hyperspectral anomaly detection to benefit from the success of the attention mechanism, which mines both local and global features for hyperspectral anomaly detection tasks [49], [50], [51]. Numerous studies have indicated that these methods perform excellently for pixel-level anomaly detection, relying on the powerful computational and feature extraction capabilities of deep learning [52], [53], [54]. However, the overfitting problem inevitably arises when detecting large anomalies. The underlying reasons are mainly two: on the one hand, anomaly detection is an unsupervised task, and large-target detection scenes lead to the fact that the network training focuses on image reconstruction rather than on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

anomaly detection. On the other hand, the network design lacks physical significance and reliability, blending information about different anomalies during the image reconstruction process.

C. Current Challenges and Weaknesses

Currently, in addition to classical statistical methods, most of the mainstream anomaly detection methods are carried out using the idea of image reconstruction. In this context, how to successfully detect anomalies while completely reconstructing the background is a challenging objective. Existing hyperspectral anomaly detection networks generally employ background guidance [41], [44], regularization constraints [37], [48], and statistical distribution assumptions [43] to suppress the reconstruction of anomalies. However, in real-world applications, due to the limitations of sensor devices and the effects of varying illumination, complex and diverse backgrounds and anomalies do not exhibit highly distinctive spectral and spatial differences. Specifically, sensor resolution makes the edges of anomalies and backgrounds unclear, while changes in illumination conditions cause spectral variability. In turn, sparse backgrounds behave consistently with anomalies in a spatial sense. At this point, anomaly detection becomes extremely difficult when feeding the hyperspectral anomaly detection network with the entire image and thus eliminating the negative impact of potential anomaly contamination on background reconstruction is essential.

Recently, the masked autoencoder (MAE) has demonstrated its ability to reconstruct complete images using a high proportion of missing data [55], providing a new avenue for hyperspectral anomaly detection network design schemes. The question is whether a small amount of background is indeed sufficient to learn an effective representation of the complete image, thus eliminating anomalies while reconstructing the background. In this respect, two fundamental issues naturally arise.

1) Since anomalous targets come in different shapes, sizes, and distributions, how to construct a valid mask to cover potential anomalies?
2) Although the covering probability of the random mask in MAE is extremely high (around 75%), the model sees almost the entire image over multiple training sessions. However, the masks of potential anomalies are fixed. How to ensure a high-quality reconstruction of the background in this case?

With these two concerns in mind, in this work, the core idea of MAE is introduced into fully convolutional networks, developing a new sparse-inspired masked convolutional network (SMCNet) for hyperspectral anomaly detection. The novel SMCNet primarily consists of three intersupporting components: 1) a hierarchical encoder; 2) a sparse projection layer; and 3) a hierarchical decoder. With an adaptive potential anomaly masking strategy, the encoder extracts multidimensional features of the remaining background with the help of sparse convolution. Rather than the commonly used positional embedding strategy, a novel sparse-inspired projection layer is built to populate the unencoded regions for complete image recovery. In the decoder, a hybrid attention mechanism (local–middle–global and spatial–spectral) is designed to refine the background during image recovery. Different from other fully convolutional network-based hyperspectral anomaly detection methods, the design of the proposed SMCNet removes the negative impact of potential anomalies on the background reconstruction from the root, providing a simpler and more effective solution.

The main innovative contributions of this work can be summarized as follows.

1) Inspired by MAE, a new SMCNet is designed for hyperspectral anomaly detection, dedicated to learning an efficient representation of the complete background reconstruction from a small amount of background samples, which not only achieves an accurate reconstruction of the background but also highlights the salience of the anomalies.
2) To support the proposed framework, a potential anomaly masking strategy based on adaptive segmentation is designed. Meanwhile, a matching sparse convolution is employed in the encoder to guarantee accurate extraction of background features while reducing computational complexity.
3) To boost background reconstruction, a sparse-inspired projection layer and a hybrid attention mechanism are developed. The former delicately exploits background features to fill holes in the encoded map, while the latter refines the background through multiscale (local–middle–global) and multidimensional (spatial–spectral) feature mining.

The remainder of this article is organized as follows. Section II describes some related work. Section III provides a detailed introduction of the newly proposed SMCNet. Section IV introduces the experimental setup and gives an in-depth analysis of the obtained anomaly detection results. Section V concludes this article with some remarks.

II. Related Work

The proposed SMCNet method involves the principal concepts of two techniques, i.e., *fully convolutional anomaly detection* and *MAE*.

A. Fully Convolutional Anomaly Detection

The fully convolutional anomaly detection method is an AE-based method, which is mainly classified into two categories according to the specific implementation process. The former is the noise-to-image reconstruction process [36], i.e., the input is uniform noise $\mathbf{N}$ with the same size as the image $\mathbf{Y}$, and the noise map is gradually approximated to the original map by encoder and decoder operations, i.e.,

$$\begin{cases} \mathbf{Z} = f_{\text{en}}(\mathbf{N}; \mathbf{W}, \mathbf{b}) \\ \hat{\mathbf{Y}} = f_{\text{de}}(\mathbf{Z}; \hat{\mathbf{W}}, \hat{\mathbf{b}}) \end{cases} \tag{1}$$

where $\mathbf{Z}$ and $\hat{\mathbf{Y}}$ are the latent space feature and the reconstructed image, respectively. $\mathbf{W}$ and $\mathbf{b}$ are the hyperparameters of the encoder f_{en}, and $\hat{\mathbf{W}}$ and $\hat{\mathbf{b}}$ are the hyperparameters of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

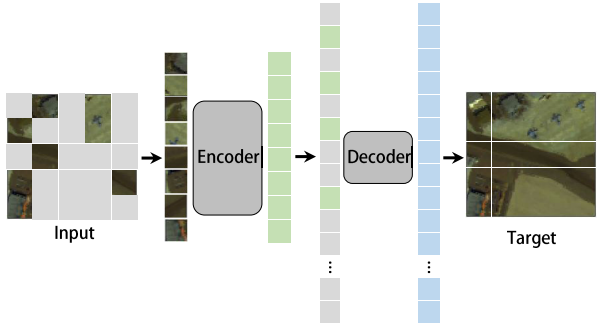


Fig. 1. Graphical illustration of the MAE. The gray patches of the input are the masked regions, which retain the position information. The encoder operates on the visible patches to obtain hidden features, i.e., green patches. Then, the hidden features and the mask tokens (gray patches) are fed into the decoder to reconstruct the image (blue patches).

the decoder f_{de}. The reconstruction residual map between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ is obtained to determine anomalies by the l_2-norm or the Mahalanobis distance.

The latter is the image-to-image reconstruction process, which differs from the first one because the encoder input is the original image, as shown in the following equation. The reconstruction is accomplished by feature squeezing and expansion operations, i.e.,

$$\mathbf{Z} = f_{\text{en}}(\mathbf{Y}; \mathbf{W}, \mathbf{b}). \tag{2}$$

Nevertheless, the crucial problem to be solved is the same for both parts, i.e., how to avoid anomaly reconstruction. Low-rank embedding [56], no reference sampling [57], loss guidance [58], and positive sample generation [59] provide limited mitigation of anomaly reconstruction, requiring the exploration of more immediate and productive methods to remove such anomalous contamination.

B. Masked Autoencoder

MAE is a scalable learning framework in computer vision with excellent performance in self-supervised and unsupervised tasks [55]. The basic idea is that images contain a massive amount of redundant information and the original image can be reconstructed based on small portions of the image. As illustrated in Fig. 1, MAE has two main designs, i.e., an asymmetric coding structure (based on the transformer framework) and a high percentage of mask rate. Specifically, a high percentage of masking operations are executed on the image (e.g., 75%), and the encoder operates only on a subset of the visible patches. Meanwhile, a lightweight decoder is deployed to reconstruct the original image from the latent representations and mask tokens.

Typically, after the MAE training is accomplished, the decoder is simply replaced to migrate to various downstream tasks, such as classification, target detection, and video tracking. Given the success of MAE, its combination with CNNs has recently been investigated, giving rise to network architectures such as CovMAE [60], ConvNeXt V2 [61], and Spark [62], which provide strategies for self-supervised learning with CNNs. To the best of authors' knowledge, there are no network architectures using MAEs for anomaly detection purposes.

III. Methodology

Considering the particularities of hyperspectral anomaly detection, a new SMCNet is developed to efficiently reconstruct the background and detect anomalies. This section provides further details on the problem statement, network deployment, and loss function.

A. Problem Statement

The goal of fully convolutional-based hyperspectral anomaly detection is to reconstruct the background and detect anomalies by residuals. Inspired by MAE, our goal is to mask out potential anomalies and rely on a small amount of background samples to reconstruct the image. Let $\mathbf{Y} \in \mathbb{R}^{H\times W\times B}$ be the hyperspectral image, where H, W, and B are the height, width, and the number of bands, respectively. The above idea, then, can be concretely translated into the following four formulas:

$$\begin{cases} \mathbf{Y}^* = \text{Mask}(\mathbf{Y}) \\ \mathbf{Z}^* = f_{\text{en}}(\mathbf{Y}^*_{\text{unmasked}}; \mathbf{W}, \mathbf{b}) \\ \mathbf{Z} = \mathbf{Z}^* + f(\mathbf{Y}^*_{\text{masked}}) \\ \hat{\mathbf{Y}} = f_{\text{de}}(\mathbf{Z}; \hat{\mathbf{W}}\hat{\mathbf{b}}) \end{cases} \tag{3}$$

corresponding to the four operations of masking, encoding, embedding, and decoding, respectively. Here, masked $\mathbf{Y}$ is $\mathbf{Y}^*$, composed by $\mathbf{Y}^*_{\text{unmasked}}$ and $\mathbf{Y}^*_{\text{masked}}$, which corresponds to the green and gray areas in Fig. 1. Accordingly, $\mathbf{Z}^*$ is the hidden space feature of $\mathbf{Y}^*_{\text{unmasked}}$. Before decoding, the function f embeds the masked region (combined with $\mathbf{Z}^*$) into $\mathbf{Z}$. Although this implementation has been formulated and specified, the following issues need to be considered at each stage for hyperspectral anomaly detection purposes.

1) *Masking:* In the proposed method, anomalies in the hyperspectral image are expected to be covered, and thus, the traditional random masking strategy is not applicable. Besides, the shape, size, and distribution of the anomalous targets are diverse and unknown. Therefore, a question arises: how to mask the anomalies accurately?
2) *Encoding:* After masking the hyperspectral image, the encoder is desired to operate on unmasked regions for background feature extraction. However, the input to the fully CNN is a regular image, which will inevitably waste computation and affect accuracy if the mask portion is filled with zeros and regular convolution is performed. Therefore, a question arises: how to extract background features efficiently?
3) *Embedding:* To avoid introducing anomalous information in the encoder, the masking strategy occludes part of the background. If conventional positional embedding is used to populate the mask region for the decoder, the background false alarm in the mask region will most probably be caused. Hence, it is necessary to contemplate embedding methods favoring background recovery.
4) *Decoding:* In any framework associated with MAE, despite the high percentage of masking strategies employed, the network tends to sense the full image information as the number of training sessions increases.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

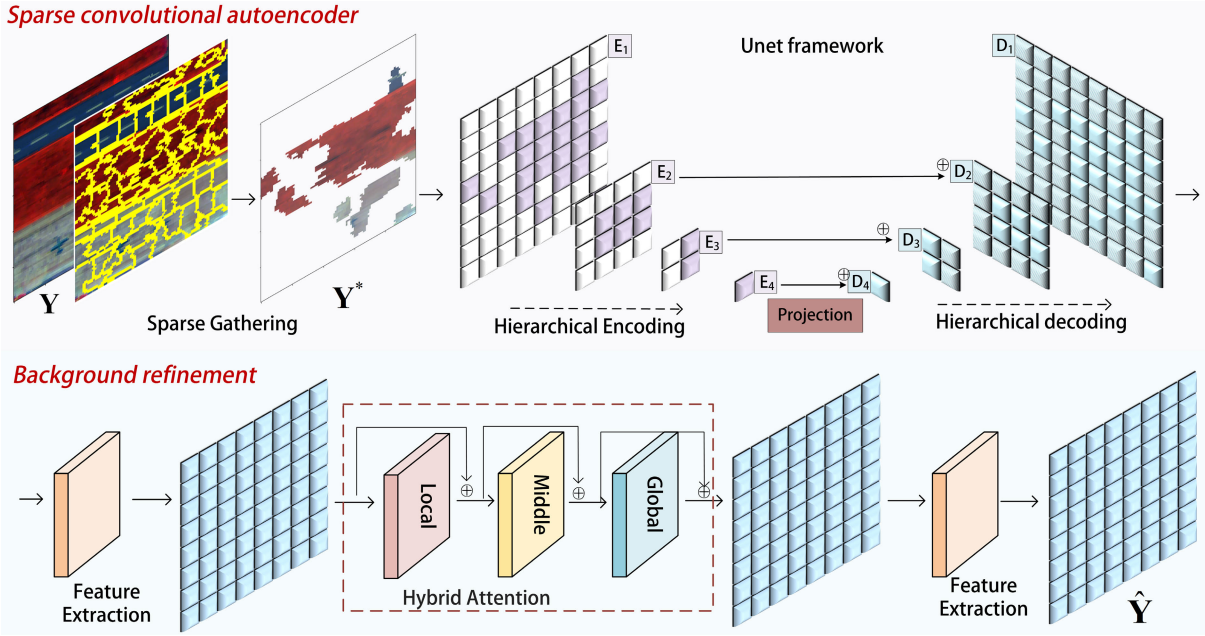


Fig. 2. Architecture of the newly proposed SMCNet, which works to reconstruct the background of the image so that anomalies can be detected by the residual map. First, the input image $\mathbf{Y}$ is gathered sparsely using superpixel segmentation and (4)–(6) to retain reliable background information. Then, the masked image $\mathbf{Y}^*$ is fed into a Unet framework, as shown in Fig. 4, in which the encoding stage is implemented by employing sparse convolution of the ME library to extract hierarchical features of backgrounds $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$, and $\mathbf{E}_4$. Before decoding, a projection strategy is devised to complement the masked region of the hidden feature map $\mathbf{E}_4$ to generate $\mathbf{D}_4$. During the decoding stage, $\mathbf{D}_4$, regular 2-D CNN, and skip connections are used to obtain the decoding feature maps $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$, in which $\mathbf{D}_1$ is the preliminary background reconstruction map with low feature dimensions. After that, a background refinement module composed of two feature extraction layers and a hybrid attention mechanism is deployed, as shown in Fig. 5, to recover the background $\hat{\mathbf{Y}}$ completely.

However, the masking strategy is static in the proposed approach, which undoubtedly poses a challenge to the global background reconstruction. As a result, it is vital to ascertain how to refine the background at the decoding stage.

B. Network Deployment

With the above considerations in mind, the proposed SMCNet is deployed using the following parts: 1) adaptive segmentation-based masking; 2) hierarchical sparse convolutional AE; 3) sparse-inspired projection layer; and 4) hybrid attention mechanism for background refinement. The overall network architecture is shown in Fig. 2.

1) Adaptive Segmentation-Based Masking: The masking strategy intended for the hyperspectral anomaly detection task consists of two stages: 1) to adaptively segment the image into compact regions with similar features and 2) to identify potential anomalous regions and mask them out. Among the adaptive segmentation algorithms, superpixel segmentation exhibits superior flexibility and applicability in image processing as a preprocessing procedure. Therefore, in this section, a superpixel segmentation method (simple linear iterative clustering) is employed to segment the hyperspectral image $\mathbf{Y}$ into $\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_n$ compact regions, where n is the number of superpixels.

For hyperspectral images containing anomalies, after segmentation, the number of background superpixels is dominant and highly correlated. Furthermore, the number of superpixels containing anomalies is small and sparsely-distributed. Before masking, the following two cases regarding the size of anomalies are considered: (*case 1*) the size of the anomaly is far smaller than the superpixel and occupies a small portion of it and (*case 2*) the size of the anomaly is greater than or equal to the superpixel, occupying the entire superpixel or even contaminating neighboring superpixels. Regarding the former case, since the spectral distribution of the anomaly is significantly different from that of the surrounding backgrounds, the spectral standard deviation of all pixels of a superpixel in case 1 is intuitively larger than that of the background superpixel, while the spectral mean within the superpixel in case 2 is intuitively far from that of the background superpixel. Depending on this, an anomalous superpixel can be masked out as follows:

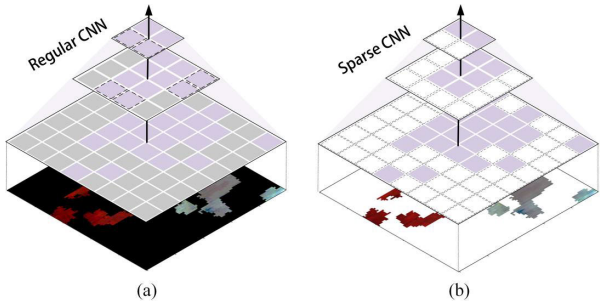


Fig. 3. (a) Regular CNN and (b) sparse CNN for a "holey hyperspectral image" with "empty parts." The mask regions of the sparse CNN are filled as blank areas, while the mask regions in the regular CNN are filled as gray areas to indicate the zero padding. The regular CNN will change the feature distribution, i.e., the black dashed regions in (a), with the increasing of convolutional layers, while the sparse CNN will keep the original feature distribution.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

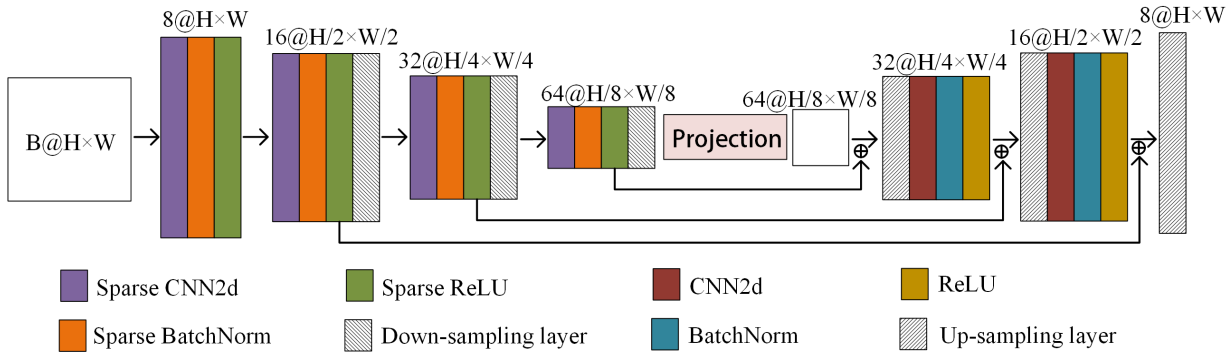


Fig. 4. Structure of the Unet framework by using sparse and regular convolution.

$$\begin{cases} \mathbf{M}(\text{where}(\mathbf{s}_i)) = 1, & \text{case 1 or case 2} \\ \mathbf{M}(\text{where}(\mathbf{s}_i)) = 0, & \text{otherwise} \end{cases} \tag{4}$$

where

$$\begin{cases} \text{case 1}, & \text{std}(\mathbf{s}_i) \in \text{top}_{\frac{n}{2}}\left(\text{sort}_{\text{descend}}(\text{std}(\mathbf{s}_j)_{j=1}^{n})\right) \\ \text{case 2}, & \text{mean}(\mathbf{s}_i) - \dfrac{1}{n}\displaystyle\sum_{j=1}^{n}\text{mean}(\mathbf{s}_j) \geq \text{std}\left(\text{mean}(\mathbf{s}_j)_{j=1}^{n}\right) \end{cases} \tag{5}$$

and $\mathbf{M}$ is the mask tensor with the same size as $\mathbf{Y}$. Here, the locations in $\mathbf{M}$ equal to 1 denote the masked region, and the masked image $\mathbf{Y}^*$ can be obtained as follows:

$$\mathbf{Y}^* = \mathbf{Y} \circ (\mathbf{1} - \mathbf{M}) \tag{6}$$

where $\circ$ is the Hadamard product, and $\mathbf{1}$ is the tensor where all elements are 1 with the same size to $\mathbf{M}$. In this case, the pixels with zeros of $\mathbf{Y}^*$ are the masked regions and others are the unmasked regions. Fig. 2 depicts the specific operation of masking, where the blank parts are the obscured areas.

2) Hierarchical Sparse Convolutional AE: Notwithstanding its ability to handle irregular inputs (with variable lengths), the original MAE is not suitable for irregularly shaped superpixels, leaving the fully convolutional structure as a natural alternative. In addition, hierarchical structures have been recognized as fundamental and necessary in computer vision tasks, allowing the network to handle objects with different sizes by extracting scale-invariant features, especially for target detection purposes. Exploiting this idea, a hierarchical structure is introduced into the fully convolutional encoder and decoder to extract multiscale background features.

Nevertheless, as discussed in Section III-A, popular convolution methods such as regular convolution, dilated convolution, and deformable convolution do not adapt to a "holey image" structure. Fortunately, a 3-D "holey hyperspectral image" is extremely similar to a sparse point cloud in 3-D tasks, where well-established sparse convolutions such as the Minkowski engine (ME) framework [63] have been developed for mining features of visible data. Inspired by this, the standard convolution in the encoder is replaced by a sparse convolution in ME as well as the corresponding normalization and activation layers. As shown in Fig. 3, the input of the regular CNN is the "holey hyperspectral image" with zero padding to keep the regular input in size. The input of the sparse CNN is only the unmasked regions of the "holey hyperspectral image," which is irregular in size. In this case, the regular CNN with zero padding not only interferes with the distribution of feature values but also extracts all the features covered by the convolution kernel, resulting in an inflated feature map. Sparse convolution, on the other hand, skips the masked regions in the hole feature map and focuses on the unmasked regions, thus preserving the background distribution and ratio in the input image with an efficient computation. The entire sparse MAE is structured using a hierarchical layout, as shown in Fig. 4, where the encoder is implemented using the sparse convolution of the ME library, and the decoder is implemented using regular convolution. Specifically, the encoder consists of a shallow feature extraction layer $\mathcal{E}_1$ and three downsampling feature extraction layers $\mathcal{E}_i, i \in \{2, 3, 4\}$

$$\begin{cases} \mathbf{E}_1 = \mathcal{E}_1(\mathbf{Y}^*), & \text{if } i = 1 \\ \mathbf{E}_i = \mathcal{E}_i(\mathbf{E}_{i-1}), & \text{if } i \in \{2, 3, 4\}. \end{cases} \tag{7}$$

After (7), the spatial resolutions of the encoded features $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$, and $\mathbf{E}_4$ are $H \times W$, $(H/2) \times (W/2)$, $(H/4) \times (W/4)$, and $(H/8) \times (W/8)$, and the corresponding feature dimensions are 8, 16, 32, and 64, respectively.

In the decoder, except for the upsampling hierarchy (symmetric to the encoder), like the Unet [64], skip connections are inserted between the encoder and the decoder, transmitting the background feature information extracted by the encoder directly to the decoder

$$\begin{cases} \mathbf{D}_i = \mathcal{P}(\mathbf{E}_i), & \text{if } i = 4 \\ \mathbf{D}_i = \mathcal{D}_{i+1}(\mathbf{D}_{i+1} + \mathbf{E}_{i+1}), & \text{if } i \in \{3, 2, 1\} \end{cases} \tag{8}$$

where $\mathcal{P}$ is a projection layer for filling in holes in sparse features as specified in Section III-B3 and $\mathcal{D}_i$ is the decoder layer with regular convolution for handling dense feature maps. Similar to the encoder, the spatial resolutions of the decoded features $\mathbf{D}_4$, $\mathbf{D}_3$, $\mathbf{D}_2$, and $\mathbf{D}_1$ are $(H/8) \times (W/8)$, $(H/4) \times (W/4)$, $(H/2) \times (W/2)$, and $H \times W$, respectively. It is noteworthy that the final output of the decoder is $\mathbf{D}_1$, with a spatial size of $H \times W$ and a feature dimension of 8. The background refinement module in Section III-B4 will further restore $\mathbf{D}_1$ to the size of the input image to complete the reconstruction, i.e., $\hat{\mathbf{Y}}$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

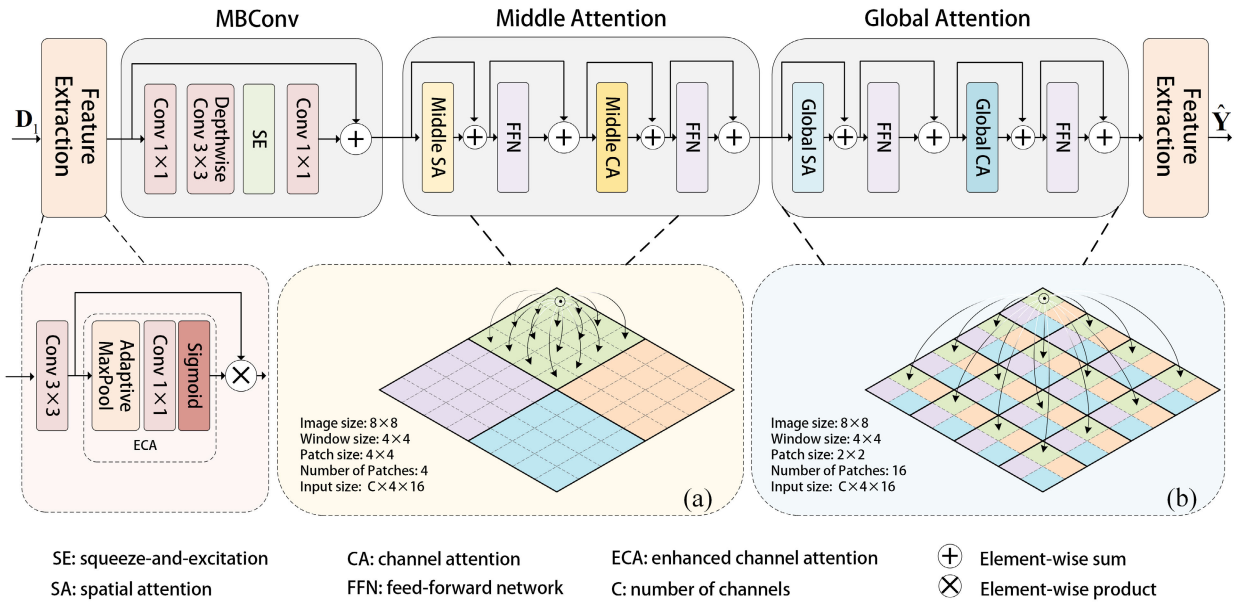


Fig. 5. Background refinement framework with local–middle–global hybrid attention. (a) and (b) Simple examples of generating patches for middle and global attentions, respectively.

3) Sparse-Inspired Projection Layer: Between the encoder of sparse convolution and the decoder of dense convolution, the projection layer is set up to tackle the mask region to generate dense features. In previous MAE-inspired networks, positional embedding is conventionally used to fill in the holes in the sparse feature maps as a straightforward and convenient method. Moreover, most of these networks are solely trained by the random masking strategy to extract desirable features for various downstream tasks, but not to reconstruct the image. In contrast, the proposed SMCNet: 1) dedicates itself to image reconstruction and focuses on the background rather than anomalies; 2) designs a masking strategy for the hyperspectral anomaly detection task, i.e., covering potential anomalies as much as possible; and 3) employs a static masking strategy rather than a dynamic one, resulting in invariable input features. This implies that the projection layer and the decoder are crucial for the anomaly detection performance, supporting both the downstream and the reconstruction tasks.

Given the complexity of the application scenario, the masked regions inevitably cover up some background pixels while obscuring potential anomalies. Therefore, at the projection layer, a differential approach to the background and anomalies of the masked region is necessary instead of using positional embedding, which is prone to cause false alarms. Because the similarity between natural objects is higher than that between natural and artificial objects, a sparse-inspired projection layer is designed motivated by the traditional representation learning-based hyperspectral anomaly detection method. Hence, the effective representation of the background extracted by the encoder can be considered as a dictionary, while the background of the masked region can be approximated by this dictionary. At the same time, the background underlying the anomalous location is approximated without introducing anomalous information. In our implementation, a *linear* layer is adopted to finish the projection of $\mathbf{E}_4$ to $\mathbf{D}_4$. Let $\mathbf{E}'_4$ be the 2-D unfolding of $\mathbf{E}_4$ with the size of $64 \times n_m$ and $\mathbf{D}'_4$ is the 2-D unfolding of $\mathbf{D}_4$ with the size of $64 \times ((H/8) \times (W/8))$, where n_m is the number of feature vectors after masking, that is,

$$\mathbf{D}'_4 = \text{linear}\left(\mathbf{E}'_4; \mathbf{W}_p\right) \tag{9}$$

where $\mathbf{W}_p$ is the projection matrix with the size of $n_m \times ((H/8) \times (W/8))$. Through (9), the masked region will be embedded with background information, which encourages the recovery of the background as well as the identification of anomalies, which is optimally applicable to hyperspectral anomaly detection tasks.

However, despite the fact that the downsampling and masking strategies filter out effective background features, this magnitude is still sizable and completely outperforms the overcomplete dictionary. According to the golden cornerstone in sparse representation, a small number of bases can achieve a compact and efficient representation of the data and promote model robustness. Besides, research has shown that complex neural network structures not only bring in numerous parameters to optimize but are also prone to over-fitting, whereas model pruning can effectively improve the inference speed and the generalization ability of the model. Therefore, in the projection layer, we adopt a sparse-inspired regularization to constrain the projection matrix as follows:

$$\mathbf{D}'_4 = \text{linear}\left(\mathbf{E}'_4; \mathbf{W}_p\right) \text{ s.t } \min \left\|\mathbf{W}_p\right\|_l \tag{10}$$

where l can be the Frobenius norm or the l_1-norm of a matrix. During the code implementation, the fully connected layer is employed to complete the projection from $\mathbf{E}'_4$ to $\mathbf{D}'_4$, and $\mathbf{W}_p$ is the corresponding weight matrix. Subtly, the projection layer can be understood as a representation process in a sparse representation model, where $\mathbf{E}'_4$ can be considered as a dictionary extracted from the background and $\mathbf{D}'_4$ is a linear

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

combination of the features in $\mathbf{E}_4'$ with the corresponding representation coefficient $\mathbf{W}_p$. To this end, the regularization constraint to $\mathbf{W}_p$ will make the representation stable and reliable, and $\mathbf{D}_4'$ only contains the background information.

4) Hybrid Attention Mechanism for Background Refinement: With respect to the popular transformer mechanism, fully convolutional networks ignore spatial and channel interactions, which are valuable for further background reconstruction in hyperspectral anomaly detection. An interesting improvement is a hybrid design using a transformer and convolutional layers, either explicitly or implicitly [65], [66]. To this end, we combine convolutional layers and hybrid attention mechanisms in the decoding phase to refine the background features to generate $\hat{\mathbf{Y}}$ from $\mathbf{D}_1$

$$\hat{\mathbf{Y}} = \mathcal{BR}(\mathbf{D}_1) \tag{11}$$

where $\mathcal{BR}$ is the background refinement module. As shown in Fig. 5, the designed background refinement module includes two feature extractors, a local convolution block, a middle attention block, and a global attention block, which endows the network with full-scale (local/middle/global scale) feature extraction capability to mine richer information for background refinement with higher information entropy. Among them, the feature extraction layer is composed of a 3×3 convolution kernel and an enhanced channel attention module. Local convolution is implemented by the lightweight MBConv module [67], while middle attention and global attention are implemented in a subtle way.

As shown in Fig. 5, middle attention works to compute attention scores between pixels within a patch, while global attention works to compute attention scores between pixels located at the same position within all patches. Although the attention domains of middle and global attentions are completely different, they are realized under the same window size. Specifically, for $H \times W$ feature maps with the window size of $P \times P$, middle attention divides the image into $(H/P) \times (W/P)$ nonoverlapping patches with a spatial size of $P \times P$. In the global attention module (without any extra parameters), the image is divided into $P \times P$ nonoverlapping patches with a spatial size of $(H/P) \times (W/P)$. In other words, the window size in middle attention is the spatial size of the patch to be divided, while the window size in global attention is the number of patches to be divided in the row and column. Even though the sizes of their input features are scaled to $C \times ((H/P) \times (W/P)) \times (P \times P)$ in the attention map computation, the implied meanings are quite different, leading to different levels of attention. Take the image with a size of 8×8 and a window size of 4×4 as an example, as shown in Fig. 5(a) and (b). In the middle attention module, the image is divided into four patches with a spatial size of 4×4, and the attention maps between pixels in a patch are computed. In the global attention module, the image is divided into 16 patches with a spatial size of 2×2, and thus, the attention maps between pixels at the same position in all the patches are computed. In brief, both can be realized by splitting the input features into nonoverlapping patches under the same parameters, which are only arranged differently in the transformer computation, i.e., leading to two scales of attention modules.

Meanwhile, to enhance channel and spatial interactions and transmit spatial and channel information in a tight mode, a spatial-channel attention module is constructed for middle attention and global attention, simultaneously, thus mining the comprehensive spatial and channel information. As shown in Fig. 6, on the spatial interaction, the input feature $\mathbf{F}_{\text{in}} \in \mathbb{R}^{HW \times C}$ is first linearly projected three times to obtain the query, key, and value metrics $\mathbf{Q}_s \in \mathbb{R}^{HW \times C}$, $\mathbf{K}_s \in \mathbb{R}^{HW \times C}$, and $\mathbf{V}_s \in \mathbb{R}^{HW \times C}$, respectively. Then, the dot product of $\mathbf{Q}_s$ and $\mathbf{K}_s$ is computed and normalized to obtain a spatial attention map of size $HW \times HW$. Finally, the spatial interaction results are obtained by weighted summation of the spatial attention map and $\mathbf{V}_s$ as follows:

$$\begin{cases} \mathbf{Q}_s = \mathcal{L}_q(\mathbf{F}_{\text{in}}), \mathbf{K}_s = \mathcal{L}_k(\mathbf{F}_{\text{in}}), \mathbf{V}_s = \mathcal{L}_v(\mathbf{F}_{\text{in}}) \\ \mathbf{Y}_s = \mathcal{A}_s(\mathbf{Q}_s, \mathbf{K}_s, \mathbf{V}_s) = \text{SoftMax}(\mathbf{Q}_s \mathbf{K}_s^{\text{T}}) \cdot \mathbf{V}_s \end{cases} \tag{12}$$

where $\mathcal{L}_q$, $\mathcal{L}_k$, and $\mathcal{L}_v$ are the linear projection operations, $\mathcal{A}_s$ is the spatial attention operation, and $\mathbf{Y}_s$ is the corresponding output. On the channel interaction, $\mathbf{Q}_c \in \mathbb{R}^{C \times HW}$, $\mathbf{K}_c \in \mathbb{R}^{C \times HW}$, and $\mathbf{V}_c \in \mathbb{R}^{C \times HW}$ can be derived from two alternative means for subsequent channel attention computation, one is a a linear projection onto $\text{FFN}(\mathbf{Y}_s)$ and the other is a direct rotation of $\mathbf{Q}_s$, $\mathbf{K}_s$, and $\mathbf{V}_s$ to obtain the transpositions. Here, the first strategy is adopted. As shown in Fig. 6, the size of the channel attention map is $C \times C$ and the size of $\mathbf{Y}_c$ is $C \times HW$. The spatial-channel interaction $\mathbf{Y}_{sc}$ can be obtained by counter-rotating $\mathbf{Y}_c$ and adding it to the spatial interaction result as follows:

$$\begin{cases} \hat{\mathbf{Y}}_s = \text{FFN}(\mathbf{Y}_s) \\ \mathbf{Q}_c = \mathcal{L}_q(\hat{\mathbf{Y}}_s), \mathbf{K}_c = \mathcal{L}_k(\hat{\mathbf{Y}}_s), \mathbf{V}_c = \mathcal{L}_v(\hat{\mathbf{Y}}_s) \\ \mathbf{Y}_c = \mathcal{A}_c(\mathbf{Q}_c, \mathbf{K}_c, \mathbf{V}_c) = \text{SoftMax}(\mathbf{Q}_c \mathbf{K}_c^{\text{T}}) \cdot \mathbf{V}_c \\ \mathbf{Y}_{sc} = \mathcal{R}_{-1}(\mathbf{Y}_c) + \hat{\mathbf{Y}}_s \end{cases} \tag{13}$$

where $\mathcal{R}_{-1}$ is the inverse rotation, $\mathcal{A}_c$ is the spectral attention operation, and FFN is a feed-forward network, as shown in Fig. 5. Briefly, we compute the attention maps corresponding to the spatial and channel orientations through sequential matrix operations and rotation matrix operations, in which spectral–spatial information integration is elegantly achieved.

C. Loss Function

In existing methods, the mean square error or the l_1-norm between $\mathbf{Y}$ and $\hat{\mathbf{Y}}$ is commonly taken as a loss function to guide the network training. From a physical meaning perspective, anomalies are sparse and noise is uniformly distributed in images, so the l_1-norm is more suitable for anomaly detection tasks

$$\mathcal{L} = \left\| \mathbf{Y} - \hat{\mathbf{Y}} \right\|_1. \tag{14}$$

Moreover, inspired by the idea of sparsity, in the projection layer, we impose sparse constraints on the weight parameters of the projection layer; thus, the loss function can be finally expressed as

$$\mathcal{L} = \left\| \mathbf{Y} - \hat{\mathbf{Y}} \right\|_1 + \lambda \left\| \mathbf{W}_p \right\|_F \tag{15}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

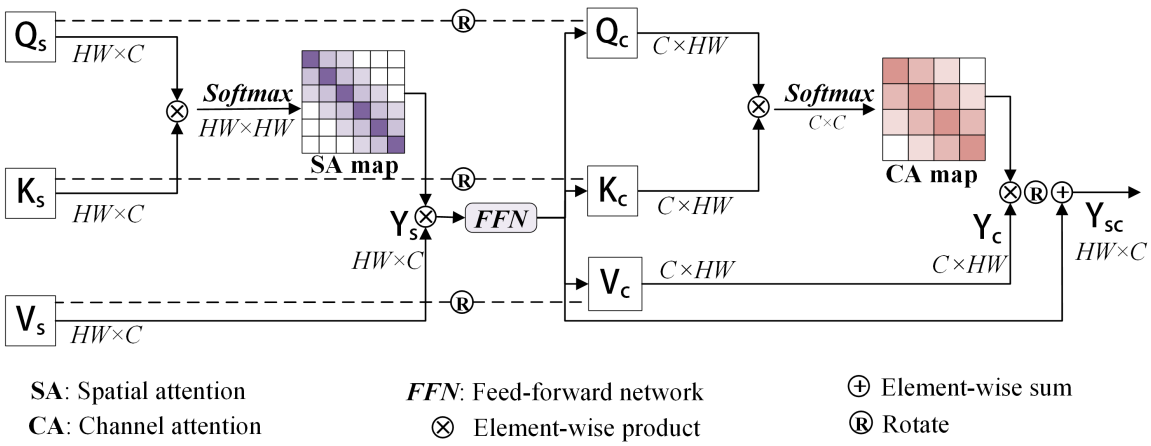


Fig. 6. Graphical illustration of spatial-channel attention.

where λ is a tradeoff parameter. After the training of the network, the anomaly map can be computed as follows:

$$\mathbf{E} = \left\| (\mathbf{Y} - \hat{\mathbf{Y}})_{:,i} \right\|_2 = \sqrt{\sum\nolimits_j (\mathbf{Y} - \hat{\mathbf{Y}})_{j,i}^2}. \tag{16}$$

The detailed pseudo-code of SMCNet is summarized in Algorithm 1.

Algorithm 1 Pseudo-Code of SMCNet

Input:

Hyperspectral dataset: $\mathbf{Y}$

Parameters: λ

Initialization: $lr = 1e-2$, $wd = 1e-6$

1: Sparse gathering by superpixel segmentation and Eqs. (4-6) to obtain $\mathbf{Y}^*$
2: **for** each epoch **do**
3: Extract encoded hierarchical feature maps $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$, and $\mathbf{E}_4$ by Eq. (7)
4: Complement the masked region of $\mathbf{E}_4$ to generate $\mathbf{D}_4$ by Eq. (10)
5: Recover hierarchical feature maps $\mathbf{D}_3$, $\mathbf{D}_2$, and $\mathbf{D}_1$ by Eq. (8)
6: Refine background by Eq. (11)
7: Train the network by minimizing Eq. (15)
8: **end for**
9: Save the `SMCNet` model with the optimal parameters
10: Run 3-6 with the above optimal parameters to obtain $\hat{\mathbf{Y}}$
11: Compute the anomaly map $\mathbf{E}$ by Eq. (16)

Output:

Anomaly map matrix: $\mathbf{E}$

IV. Experimental Results and Analysis

In this section, comprehensive experiments are conducted on six hyperspectral anomaly detection datasets using ten comparison methods to evaluate the performance of the proposed SMCNet. The details of the datasets, evaluation criteria, comparison methods and parameter setting, ablation study, and anomaly detection performances are as follows.

A. Datasets

Hyperspectral anomaly detection performance is closely linked to the characterization of the dataset. For this reason, six different types of data are selected in this study, including subpixel anomalies, small anomalies, and planar anomalies. Furthermore, these datasets involve both small and large scenes. A brief summary of the selected datasets is shown in Table I, and the details are given as follows.

1) *HYDICE Urban:* The first selected dataset was taken by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over an urban area in August 1995, and it contains a total of 80 × 100 pixels with 175 effective spectral bands. The spatial resolution is 1.56 m and the spectral resolution is 10 nm, where the anomalous target roofs and automobiles are presented at the subpixel and pixel level.
2) *ABU-B4:* The second selected dataset was taken by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor with a spatial resolution of 1.3 m over the city center of Pavia, comprising a total of 150×150 pixels and 102 effective spectral bands. As shown in the figure of Table I, the selected subimage is from the ABU dataset in [68], in which the background mainly consists of a river, a bridge, vegetation, and bare ground, where the vehicles on the bridges are regarded as small anomalous targets.
3) *San Diego I and II:* The third and fourth selected datasets come from two different subimages of the same hyperspectral image. They were collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor with a spatial resolution of 3.5 m over the San Diego airport in CA, USA, and both images consist of 100 × 100 pixels with 186 effective spectral bands. The anomalous targets in both images are three airplanes, but the difference is that the airplanes in San Diego I are smaller than those in San Diego II.
4) *Gulfport:* The fifth selected dataset was taken by the AVIRIS sensor on July 7, 2010, over the airport in Gulfport, MS, USA. It contains 100 × 100 pixels and 191 effective spectral bands, and has a spatial resolution

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

TABLE I
BRIEF SUMMARY OF THE SELECTED DATASETS. HERE, "SPA" REFERS TO THE SPATIAL RESOLUTION AND "SPE" REFERS TO THE SPECTRAL RESOLUTION

No.	Name	Pseudocolor image	Reference map	Sensor	Resolution	Size	Anomalies level
I	HYDICE Urban			HYDICE	Spa: 1.56 m Spe: 10 nm	Spa: 80×100 Spe: 175	pixel
II	ABU-B4			ROSIS-03	Spa: 1.3 m	Spa: 150×150 Spe: 102	small
III	San Diego I			AVIRIS	Spa: 3.5 m	Spa: 100×100 Spe: 186	small
IV	San Diego II			AVIRIS	Spa: 3.5 m	Spa: 100×100 Spe: 186	planar
V	Gulfport			AVIRIS	Spa: 3.4 m	Spa: 100×100 Spe: 191	planar and small
VI	CRi			Nuance Cri	Spe: 10 nm	Spa: 400×400 Spe: 191	planar

of 3.4 m. The anomalous targets are similar to those in the San Diego scenes, with three airplanes, one large and two small airplanes [68].

5) *CRi:* The sixth selected dataset was taken by the Nuance Cri sensor, with a spectral resolution of 10 nm and 46 effective spectral bands. Unlike the first five datasets, this image is larger and contains 400×400 image pixels. Relatively speaking, this image has a simple ground object composition with only grass and ten anomalous target rocks.

B. Evaluation Criteria

Besides the anomaly detection maps, we employed the receiver operating characteristic (*ROC*) curve, the area under the curve (*AUC*), and the *box plot* to assess the anomaly detection performance of all the considered methods.

The ROC curve is a common tool for evaluating the performance of a target detection model, which draws the detector's performance at different thresholds with the true positive rate (TPR) as the vertical coordinate and the false positive rate (FPR) as the horizontal coordinate. In other words, the TPR represents the proportion of correctly detected anomalies to all anomalous samples, while the FPR denotes the ratio of the number of background samples incorrectly detected as anomalies to the number of all background samples, computed as follows:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TN}}, \quad \mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FN}} \tag{17}$$

where TP denotes the number of detected anomalous samples, TN denotes the total number of anomalous samples, FP denotes the number of background samples detected as anomalies, and FN denotes the total number of background samples.

The AUC is the area under the ROC curve, which is used to quantitatively evaluate the performance of the detector. The value of AUC ranges from 0 to 1, where 0.5 means a random detector, i.e., the performance of the model is comparable to that of a random guess. An AUC greater than 0.5 means that the detector's performance is superior to that of a random guess, and the closer the AUC is to 1, the better the detector's performance.

The box plot is a visualization tool for showing the data distribution. In anomaly detection, the background and anomalies in the anomaly map are usually plotted as two boxes to observe their distributions; the larger the interval between the background box and the anomaly box, the higher the separation between the anomalies and the background, and the shorter the background box indicates a better reconstruction of the background.

C. Comparison Methods and Parameter Setting

Ten competitors, including deep and nondeep methods, are considered to validate the effectiveness of the proposed network. The details are as follows.

1) Nondeep Competitors: RX [14] is the most fundamental and representative benchmark algorithm for hyperspectral anomaly detection (without any parameters). Kernel isolated forest detector (*KIFD*) [69], which performs anomaly detection by constructing isolated forests, is an emerging statistic-based hyperspectral anomaly detection method, and its experimental parameters are set as indicated by the authors. *CRD* [19] is a classical hyperspectral anomaly detection method based on representation learning. In the experiments, the dual window size for CRD ranges from {(15, 13), (15, 11), (15, 9), (15, 7), (15, 5), (15, 3)} and λ is fixed to 1e-6. *PTA* [25] is a hyperspectral anomaly detection method based on tensor decomposition; the parameters μ, α, β, and τ are set in the range of {1e-4, 1e-3, 1e-2, 1e-1, 1e0, 1e1, 1e2, 1e3, 1e4} and the rank is set below 20.

2) Deep Competitors: Robust graph AE detector (*RGAE*) [37] and guided AE detector (*GAED*) [70] are AE-based deep learning methods, where the former uses graph regularization to constrain hidden features to weaken the representation of anomalies, and the latter uses spectral similarity constraints to guide the generation of hidden features. In the experiments, the regularization tradeoff parameter of RGAE is set in the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

TABLE II
OPTIMAL PARAMETERS SETTING OF THE CONSIDERED METHODS FOR SIX DATASETS

Methods	Parameters	HYDICE Urban	ABU-B4	San Diego I	San Diego II	Gulfport	CRi
CRD	$window_size, \lambda$	$(13, 15), 10^{-3}$	$(3, 15), 1$	$(13, 15), 10^{-4}$	$(13, 15), 10^{-2}$	$(13, 15), 1$	$(9, 15), 10^{-3}$
KIFD	M, q, τ	$3\% \times N, 10^3, 300$					
PTA	$\mu, rank, \alpha$ β, τ	$10^{-4}, 1, 10^{-4}$ $1, 1$	$10^{-1}, 9, 10^{-4}$, $1, 1$	$10^{-3}, 1, 10^{-4}$ $10^{-1}, 1$	$10^{-3}, 1, 10^{-4}$ $1, 1$	$10^{-4}, 1, 10{-}1$ $10^{-2}, 10^{-1}$	$10^{-2}, 1, 1$ $10^{-2}, 10^{-4}$
RGAE	λ, S, n_hid	$10^{-4}, 150, 10^2$	$10^{-3}, 150, 10^2$	$10^{-2}, 150, 10^2$	$10^{-4}, 150, 10^2$	$10^{-2}, 150, 10^2$	$10^{-2}, 10^3, 10^2$
GAED	$window_size$	7					
PDBSNet	$lr, factor_train$ $factor_test$	$10^{-4}, 1, 1$	$10^{-4}, 3, 3$	$10^{-4}, 4, 4$	$10^{-4}, 5, 5$	$10^{-4}, 5, 5$	$10^{-5}, 1, 1$
BockNet	$lr, window_size$	$10^{-4}, 13$	$10^{-4}, 5$	$10^{-4}, 15$	$10^{-4}, 9$	$10^{-4}, 13$	$10^{-6}, 1$
SMCNet	$epoch, \lambda$	$1000, 10^{-4}$	$1000, 10^{-4}$	$1000, 1$	$1000, 1$	$1000, 1$	$500, 10^{-4}$

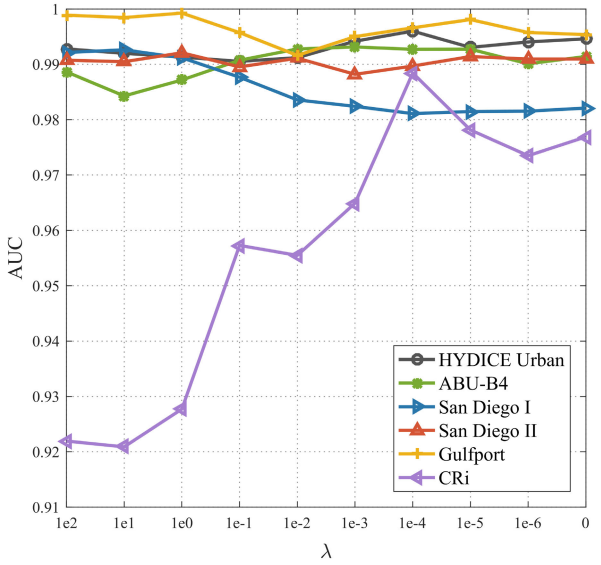


Fig. 7. Sensitivity analysis of SMCNet to parameter λ.

TABLE III
ABLATION EXPERIMENTS ON THE REGULARIZATION TERM FOR THE SIX CONSIDERED DATASETS. I–VI ARE HYDICE URBAN, ABU-B4, SAN DIEGO I, SAN DIEGO II, GULFPORT, AND CRI, RESPECTIVELY

	I	II	III	IV	V	VI
w/o $\|\mathbf{W}_p\|_F$	0.9946	0.9914	0.9820	0.9910	0.9953	0.9769
w/ $\|\mathbf{W}_p\|_F$	**0.9981**	**0.9931**	**0.9931**	**0.9927**	**0.9993**	**0.9884**

TABLE IV
COMPARATIVE RESULTS OF ABLATION EXPERIMENTS ON THE MASKING STRATEGY FOR THE SIX CONSIDERED DATASETS

Dataset	without mask	random mask	without *proj*	SMCNet
HYDICE Urban	0.9975	0.9840	0.9955	**0.9981**
ABU-B4	0.9908	0.9858	0.9801	**0.9931**
San Diego I	0.9821	0.9909	0.9687	**0.9931**
San Diego II	0.9790	0.9902	0.9869	**0.9927**
Gulfport	0.9906	**0.9993**	0.9917	**0.9993**
CRi	0.9837	0.9756	0.9540	**0.9884**
Average	0.9873	0.9876	0.9795	**0.9941**

range of {1e-4, 1e-3, 1e-2, 1e-1}. *Auto-AD* [36] is the most representative fully convolutional anomaly detection network, which achieves anomaly detection by recovering the hyperspectral image from the random noise map. *PDBSNet* [40] and *BockNet* [39] are two excellent works based on no reference-spot net, dedicated to suppressing anomalous recovery through local masking strategies. In the experiments, the factor-train and factor-test of PDBSNet are set in the range of {1, 2, 3, 4, 5, 6}, the window size of BockNet is set in the range of {1, 3, 5, 7, 9, 11, 13, 15}, and the learning rate (lr) for both PDBSNet and BockNet is set in {1e-6, 1e-5, 1e-4, 1e-3, 1e-2, 1e-1}. The gated transformer for hyperspectral anomaly detection (*GT-HAD*) [49] is the latest work that introduces the transformer into anomaly detection, enabling differential reconstruction of backgrounds and anomalies through a gating device. The parameters of GAED, Auto-AD, and GT-HAD are set to the values recommended by the authors. The optimal parameters for the considered methods are listed in Table II, except for RX, Auto-AD, and GT-HAD without tuning parameters.

3) Proposed SMCNet: In the proposed SMCNet, the trade-off parameter λ of sparse-inspired term is set in the range {1e2, 1e1, 1e0, 1e-1, 1e-2, 1e-3, 1e-4, 1e-5, 1e-6, 0}. Parametric analysis experiments are performed on all datasets using SMCNet, as shown in Fig. 7. It can be observed that, for the CRi data, the accuracy of SMCNet shows an increasing and then decreasing trend with the decrease of λ, reaching its maximum value at $\lambda = 1e-4$.

For the San Diego I data, the accuracy of SMCNet showed a slow decreasing trend with the decrease of λ and then remained stable. For the other datasets, the accuracy of SMCNet exhibits small fluctuations with the decrease of λ. In the experimental results presented in this article, λ is set to 1 for San Diego I and II and Gulfport data, and λ is set to $1e-4$ for HYDICE Urban, ABU-B4, and CRi data.

In further analysis, the setting of λ may be related to the background's complexity and the background's similarity to the anomaly in the image. The CRi data, despite the simplicity of the background, have a high similarity between the background and the target, resulting in a SMCNet performance that is sensitive to λ. In practical applications, λ can

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

take the empirical value of 1 for the scenes with complex background and $1e-4$ for the scenes with simple background or high similarity between the background and anomaly. Note that the case $\lambda = 0$ is also considered and the AUCs of SMCNet with/without the regularization constraint are listed in Table III, which indicates the effectiveness of regularization. In addition, during the optimization, the Adam optimizer is adopted, with a learning rate of 1e-2 and a weight decay of 1e-6. The number of training epochs is up to 500 (dataset VI) or 1000 (others).

Regarding the hardware and software environment, for fairness, all nondeep learning methods are implemented in the MATLAB 2016b platform on a laptop equipped with Intel Core i7 CPU and 8-GB RAM, and deep learning methods are implemented in the PyTorch framework on a laptop equipped with Intel Core i7 CPU, 32-GB RAM, and NVIDIA GeForce RTX 3070 GPUs.

D. Ablation Experiment

Three indispensable components determine the experimental performance of our newly designed SMCNet: 1) the adaptive masking strategy; 2) the sparse-inspired projection layer; and 3) the hybrid attention for the background refinement module. This section will discuss the performance and impact of these three components in detail via ablation experiments.

1) Adaptive Masking Strategy and Sparse-Inspired Projection Layer: Inspired by MAE in self-supervised deep learning, we consider adopting a masking strategy to cover potential anomalies, eliminating the possibility of anomaly contamination from the source. However, the generalized random masking strategy neglects the specificity of anomaly detection applications. Thus, we further design an adaptive masking strategy as well as a matching sparse-inspired projection layer. To verify its effectiveness, we devise four experimental schemes for all datasets using SMCNet as follows and compute the corresponding AUCs as shown in Table IV.

1) *Without Mask:* The input is the original hyperspectral image $\mathbf{Y}$. Accordingly, the hierarchical coding phase uses regular 2-D convolution instead of sparse convolution, and the projection layer is removed. The remaining network architecture is the same as that of SMCNet.
2) *Random Mask:* The original hyperspectral image is partitioned into nonoverlapping 5×5 patches. Then, following MAE's behavior, these patches are randomly masked in each epoch using a 75% masking rate. In addition, the projection layer is modified to positional embedding, and the remaining network architecture is the same as that of the proposed SMCNet.
3) *Without proj:* The sparse-inspired projection layer in SMCNet is replaced with positional embedding.

By comparing SMCNet without mask, SMCNet with random mask, and SMCNet in Table IV, it can be observed that the masking strategy is significantly effective for large-target detection scenarios (San Diego I, San Diego II, and Gulfport), where the adaptive mask outperforms the random mask. This is consistent with our intuitive reasoning that large targets are more likely to negatively affect background reconstruction, while a random masking strategy can weaken the possibility of anomaly contamination (adaptive masking can fundamentally address the problem). However, for HYDICE Urban, ABU-B4, and CRi data, the random masking strategy is detrimental. There are two main reasons. One is that, in the HYDICE Urban and ABU-B4 datasets, the interference of pixel-level anomalies on the background reconstruction is tiny, and random masking will lose more valuable background information. The other is that, in the CRi dataset, due to the high similarity between the background and anomalies, random masking weakens both anomalies and the background simultaneously. In comparison, the adaptive masking strategy can implement masks according to different anomaly detection scenarios, which has better generalization ability and competitiveness. Moreover, the comparison results between SMCNet without projection and SMCNet dramatically demonstrate the importance of the sparse-inspired projection layer for SMCNet, especially for large-target anomaly detection scenarios.

2) Hybrid Attention for Background Refinement: Our SMCNet is designed with hybrid attention in the decoder, refining the background reconstruction by extracting the delicate spatial and spectral features of the hyperspectral image. To validate the effectiveness of this module, we designed four groups of comparative experiments using SMCNet for the considered datasets as follows and calculated the corresponding AUCs (see Fig. 8).

1) *Spectral (spe) and Spatial (spa) Attention:* The former retains spectral attention both in the middle and global attention modules, while the latter retains spatial attention both in the middle and global attention modules, and the local attention module remains the same.
2) *Local, Middle, and Global:* Only one scale of attention module is employed in SMCNet, i.e., local attention module, middle attention module, or global attention module.
3) *Mix 1, Mix 2, and Mix 3:* Hybrid attention with two scales, where mix 1 is the combination of the local and the middle attention, mix 2 is the combination of local and global attention, and mix 3 is the combination of the middle and the global attention.
4) *All:* The full SMCNet.

As shown in Fig. 8, in the first group of comparison experiments, it can be observed that spatial attention is more important than spectral attention for the HYDICE Urban, Gulfport, and CRi datasets, while spectral attention is slightly more important than spatial attention for the San Diego I and II and ABU-B4 datasets. In the second and third groups of comparison experiments, it can be observed that, for different anomaly detection scenarios, different scaled attention modules contribute to background reconstruction in different degrees. Relatively speaking, for datasets with complex backgrounds (HYDICE Urban, ABU-B4, San Diego I and II, and CRi), the spatial and spectral features extracted by the middle and global attention modules are more conducive to background reconstruction. Therefore, in practical unknown and complex application scenarios, the multiscale spatial and spectral features not only improve the anomaly detection accuracy but also enhance the generalization ability.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

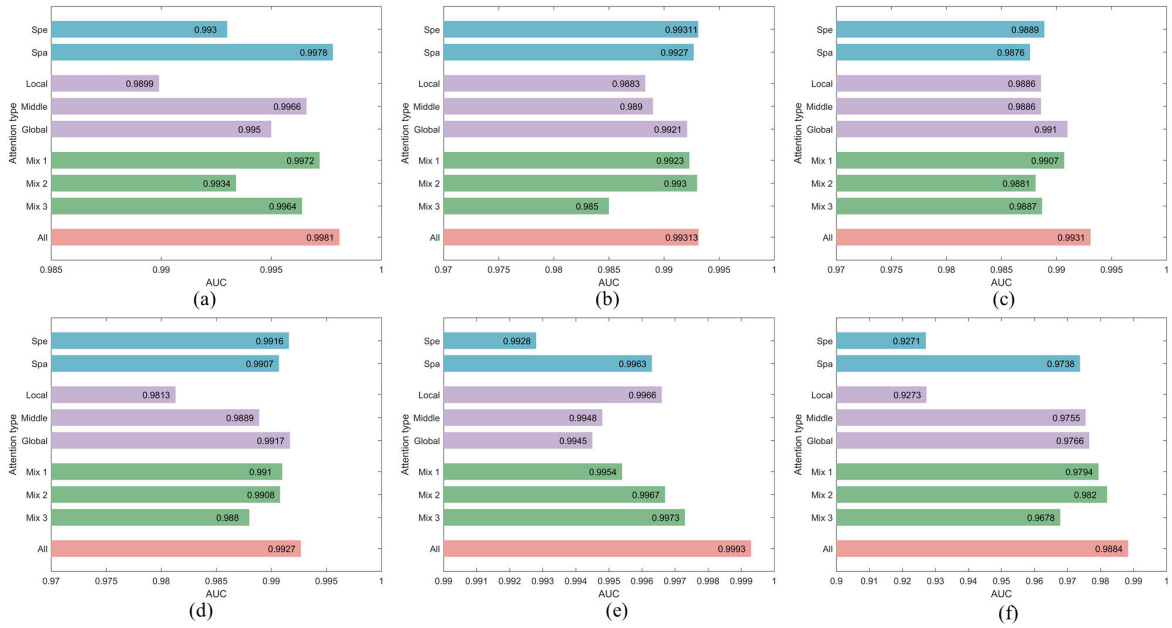


Fig. 8. Comparative results of ablation experiments on hybrid attention for the six considered datasets. (a) HYDICE Urban. (b) ABU-B4. (c) San Diego I. (d) San Diego II. (e) Gulfport. (f) CRi.

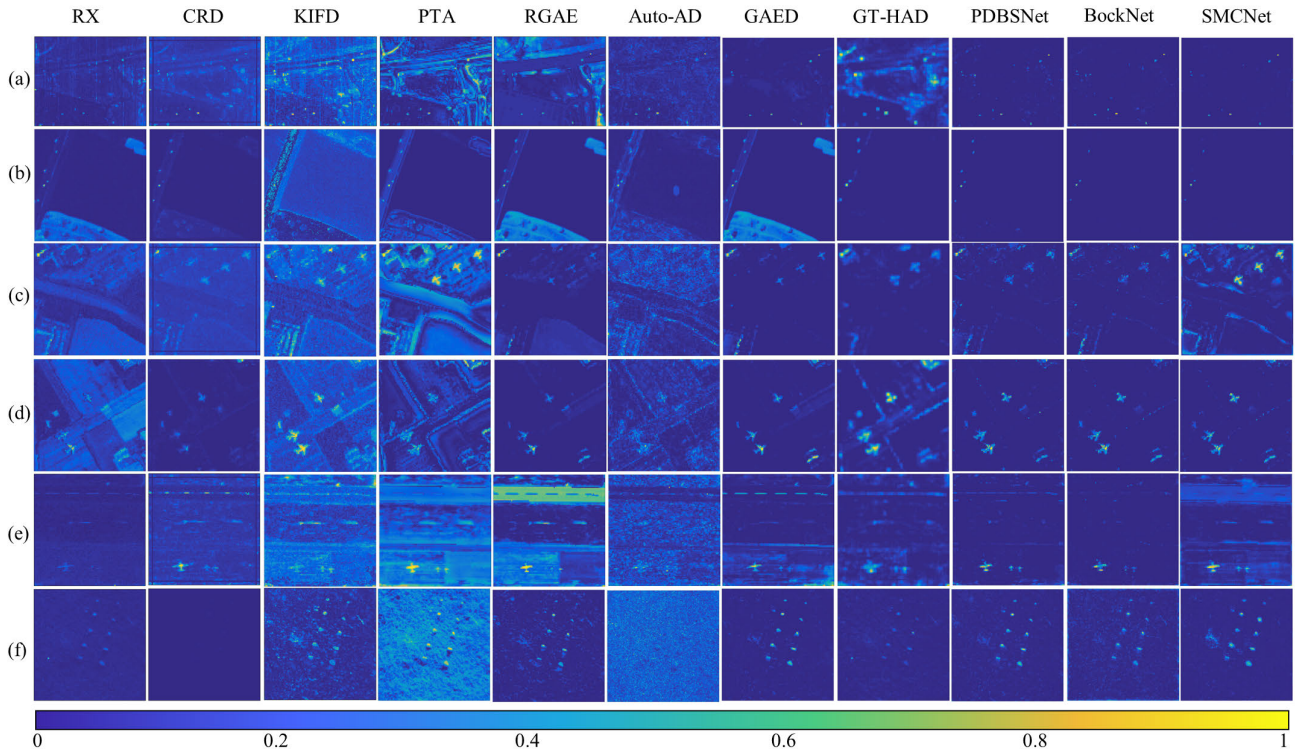


Fig. 9. Anomaly maps produced by the considered detectors on six datasets. From top to bottom: (a) HYDICE Urban; (b) ABU-B4; (c) San Diego I; (d) San Diego II; (e) Gulfport; and (f) CRi.

E. Anomaly Detection Results

In Sections IV-E1–IV-E4, the anomaly detection results obtained by the ten aforementioned detectors on the six considered datasets are discussed. First, the anomaly detection results of each detector are displayed in Fig. 9, and the box plots are shown in Fig. 10 for qualitative analysis. Then, the ROC curves are plotted in Fig. 11, and the corresponding AUC values are tabulated in Table V for quantitative evaluation, where the best and second best values are highlighted in bold and underlined formats, respectively. Along with this, a comprehensive analysis was carried out according to the results of each evaluation indicator.

1) Anomaly Maps: From a visual perspective, classical methods such as RX and CRD largely fail to highlight the accurate location of large anomalies, as shown in Fig. 9(c)–(f). This is particularly noticeable in the San Diego I

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

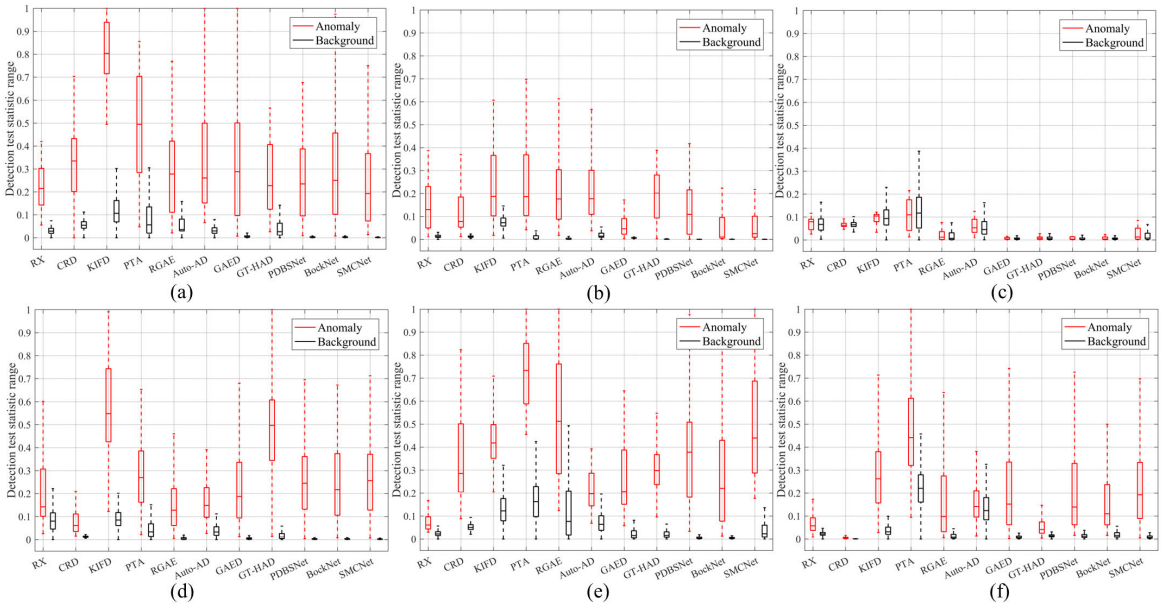


Fig. 10. Box plots of the considered detectors on six datasets: (a) HYDICE Urban; (b) ABU-B4; (c) San Diego I; (d) San Diego II; (e) Gulfport; and (f) CRi.

TABLE V
AUC AND PROCESSING TIME (S) OF THE CONSIDERED METHODS WHEN PROCESSING THE SIX CONSIDERED DATASETS

Dataset		RX	CRD	KIFD	PTA	RGAE	Auto-AD	GAED	GT-HAD	PDBSNet	BockNet	SMCNet
HYDICE Urban	AUC	0.9857	0.9499	0.9972	0.9122	0.8128	0.9939	0.9663	0.9049	0.9876	0.9907	**0.9981**
	Time (s)	0.35	17.62	68.76	49.98	150.11	50.51	68.21	10.21	71.98	70.38	42.52
ABU-B4	AUC	0.9075	0.9819	0.7965	0.9706	0.9042	0.9832	0.8968	0.9902	0.9884	0.9927	**0.9931**
	Time (s)	0.41	225.21	299.42	144.74	501.50	25.04	245.78	29.86	150.86	136.51	73.27
San Diego I	AUC	0.8545	0.9906	0.9846	0.9916	0.9742	0.8726	0.9869	0.9868	0.9891	0.9847	**0.9931**
	Time (s)	0.17	10.82	117.61	57.82	193.28	14.12	91.85	15.34	81.33	84.69	46.99
San Diego II	AUC	0.7760	0.9786	0.9917	0.9476	0.9891	0.9316	0.9884	0.9797	0.9912	0.9910	**0.9927**
	Time (s)	0.27	10.69	88.10	57.75	190.52	16.00	86.48	14.91	79.92	70.27	51.08
Gulfport	AUC	0.9526	0.9917	0.9792	0.9978	0.8890	0.9226	0.9690	0.9936	0.9974	0.9961	**0.9993**
	Time (s)	0.18	10.92	102.80	54.19	217.36	18.06	88.22	14.74	80.47	72.09	27.07
CRi	AUC	0.9135	0.9571	0.9709	0.8936	0.9109	0.5698	0.9839	0.9078	0.9821	0.9537	**0.9884**
	Time (s)	0.73	526.10	515.65	190.22	1651.31	106.16	923.24	428.95	865.93	896.72	134.81
Average	AUC	0.8983	0.9750	0.9534	0.9522	0.9134	0.8790	0.9652	0.9605	0.9893	0.9848	**0.9941**
	Time (s)	0.35	133.56	198.72	92.45	484.01	38.32	250.63	85.67	221.75	221.78	62.62

and CRi datasets. In contrast, the detection of small anomalies (HYDICE Urban and ABU-B4) all proceed normally, albeit with varying degrees of background interference. As advanced versions, the latest representatives of traditional methods (KIFD and PTA) provide superior detection of anomalies, both for large and small targets, but with slightly inferior background suppression (with KIFD exhibiting lower antinoise performance). The primary reason is that both methods implement anomaly detection with decomposition-based ideas, focusing on the prominence of anomalies and ignoring background suppression. In comparison, deep learning-driven anomaly detection methods (RGAE, GAED, PDBSNet, BockNet, GT-HAD, and SMCNet) not only emphasize the exact locations of anomalies but also suppress the background better, except for the Auto-AD. In this regard, Auto-AD is equipped with a fully convolutional structure and specializes in capturing anomalies in small-target anomaly detection scenarios (HYDICE Urban and ABU-B4); thus, it is unable to perform properly in large-target anomaly detection scenarios, as shown in Fig. 9(c)–(f), in which anomaly detection maps are full of noise. The anomaly maps of GT-HAD (motivated by the transformer) are severely patchy, which mainly arises from the fact that the method operates on nonoverlapping patches without edge processing. AE-driven RGAE and GAED work properly for most of the datasets, but the limited spatial information restricts the performance of these methods with large targets and complex scenes, which is especially obvious in HYDICE Urban, San Diego I, and Gulfport scenes. It is noteworthy to highlight that the anomaly maps of PDBSNet, BockNet, and SMCNet are similar, especially in HYDICE Urban, ABU-B4, and San Diego II datasets. This is mainly because they both use a masking strategy to suppress anomaly

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

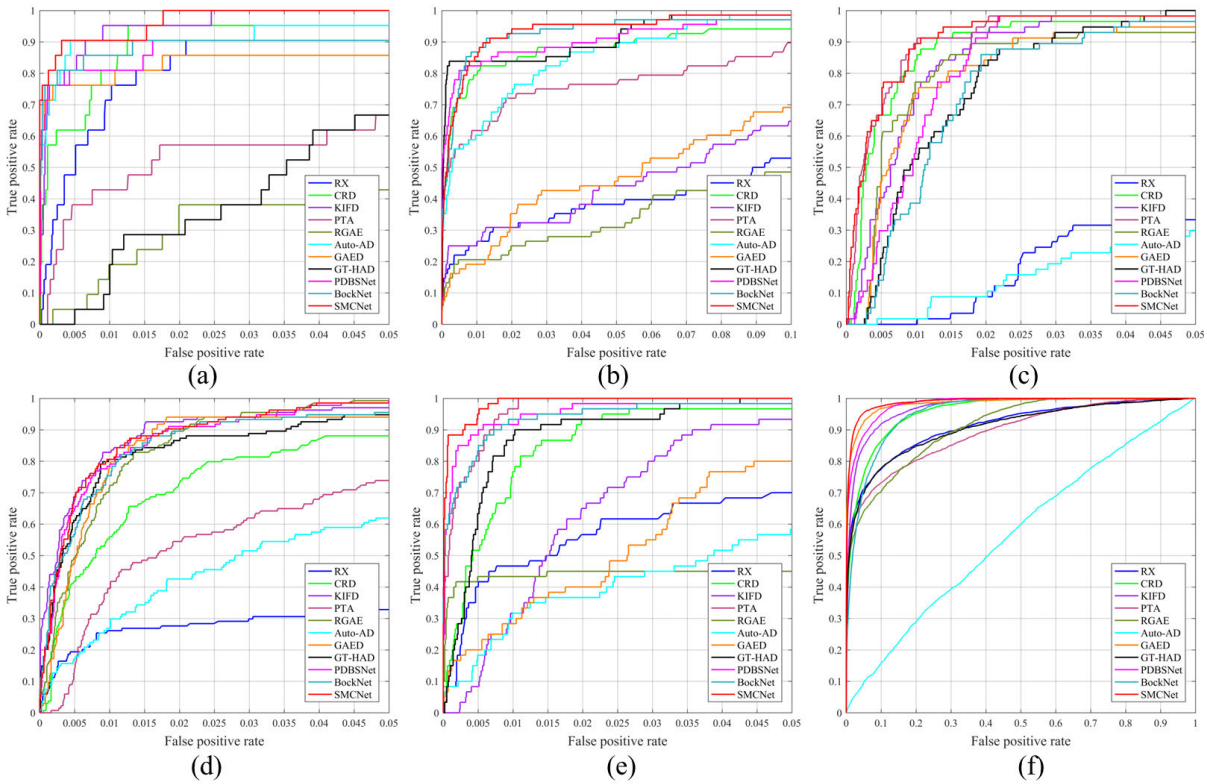


Fig. 11. ROC curves for the considered detectors on six datasets. (a) HYDICE Urban; (b) ABU-B4; (c) San Diego I; (d) San Diego II; (e) Gulfport; and (f) CRi.

recovery, the only difference is that the PDBSNet and BockNet use a local masking strategy, while the proposed SMCNet uses a global masking strategy. And the effect of the two shielding strategies is significantly different for the scenarios containing small backgrounds, as shown in Fig. 9(c), (e), and (f). PDBSNet and BockNet with local masking strategy can suppress background better, but at the same time weaken the anomalies. While SMCNet with global masking strategy can highlight the anomalies better, but is accompanied by a certain degree of false alarms. This is because the global masking strategy can easily mask small backgrounds although the background refinement module can alleviate this problem, whereas the local masking strategy has difficulty in suppressing backgrounds and highlighting anomalies at the same time due to the fixed-size parameters. In contrast, it can be seen that the proposed SMCNet is capable of accurately highlighting the locations of anomalies under the constraints of sparse convolution and sparse mapping.

2) Separation of Background and Anomalies: In the box plot, we focus on the degree of separation between background and anomalies and on the suppression degree of the background, where a method achieving both properties is considered to be optimal. From a holistic perspective, the performance of each competitor in different datasets varies significantly, which is mostly attributed to the fact that the degree of background suppression is closely related to the complexity of the background, while the separation degree of the background and the anomalies is linked to their similarity. As shown in Fig. 10, for the ABU-B4 and CRi data, most of the competitors can suppress the background satisfactorily due to the simplicity of the background. For the San Diego I and CRi data, all of the competitors are not able to distinguish the anomalies from the background accurately due to the high similarity between the background and the anomalies. In contrast, all the competitors work normally in the HYDICE Urban, ABU-B4, San Diego, and Gulfport datasets. In particular, for all considered datasets, KIFD and PTA invariably highlight anomalies and separate out the background despite their weak background suppression performance, which is compatible with the visualization of anomaly maps. Moreover, as shown in Fig. 10(b), (d), and (e), deep learning methods (RGAE, GAED, PDBSNet, BockNet, GT-HAD, and SMCNet) tend to accurately separate the anomalies from the background and suppress the background more effectively when compared to traditional methods. It is remarkable that these two aspects of the novel SMCNet method's performance are impressive among all competitors.

3) Anomaly Detection Accuracy: To emphasize the differences in the quantitative performance of each competitor, the ROC curves for each dataset appropriately scaled are provided in Fig. 11. As it can be seen, the proposed SMCNet is remarkably competitive in Gulfport and CRi datasets, consistently leading the other contenders in detection rate as the false alarm rate grows. For the other datasets, the SMCNet method performs similarly to several excellent competitors, but still outperforms them by a narrow margin, as seen from the AUCs listed in Table V. Moreover, it can be observed that the traditional method CRD, as well as the deep learning methods PDBSNet and BockNet, is more competitive than the other methods although the novel SMCNet method is still better. In terms of generalization performance over the considered datasets, the performance ranking of each competitor can be

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

established as SMCNet > PDBSNet > BockNet > CRD > GAED > GT-HAD > KIFD > PTA > RGAE > RX > Auto-AD, where the proposed SMCNet is significantly outperforming the other competitors.

4) Summary: From our qualitative and quantitative analyses, it can be concluded that CRD exhibits better anomaly detection capacity than other traditional methods. In contrast, the deep learning-driven methods achieve a superior background suppression effect, which cannot be separated from the support of their feature mining ability and powerful learning capability. Remarkably, the performance of fully convolutional Auto-AD varies significantly for small- and large-target anomaly detection scenarios, mainly because the anomalous features are not eliminated in the background reconstruction process, which is critical for large-target detection purposes. To solve this shortcoming, the designed SMCNet employs an adaptive masking strategy and a matched sparse fully convolutional network to mine the background features, and then reconstructs the image for superior anomaly identification; meanwhile, the background refinement module enhances the effective representation of the background, thus suppressing it and making the proposed SMCNet method very competitive among fully convolutional anomaly detection networks.

V. Conclusion

In this article, a new SMCNet is developed for hyperspectral anomaly detection. It consists of three mutually supportive parts: 1) a hierarchical encoder; 2) a sparse projection layer; and 3) a hierarchical decoder. Among them, the encoder extracts pure and effective background features by an adaptive anomaly masking strategy and sparse convolution. The sparse-inspired projection layer suppresses anomalies while promoting background representation by utilizing background features to fill the uncoded region. In the decoder, the designed hybrid attention mechanisms (local–middle–global and spatial–spectral) extract multiscale and multidimensional features to further refine the background. The design of SMCNet radically eliminates the negative impact of potential anomalies on the background reconstruction. Extensive experiments on six datasets show that SMCNet outperforms other state-of-the-art methods. Nonetheless, SMCNet still has an unsolved problem that the number of parameters in the projection layer computed by *linear* rises sharply with the increase of data scale, which is a potential issue for future large-scale applications. Although sparse excitation can alleviate this problem to some extent, it is not the most immediate and optimal solution. In the future, this concern will be addressed by improving the projection layer computation using dynamic weight deactivation.

References

[1] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[4] S. Matteoli, M. Diani, and G. Corsini, "A tutorial overview of anomaly detection in hyperspectral images," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 25, no. 7, pp. 5–28, Jul. 2010.

[5] B. Du, R. Zhao, L. Zhang, and L. Zhang, "A spectral–spatial based local summation anomaly detection method for hyperspectral images," *Signal Process.*, vol. 124, pp. 115–131, Jul. 2016.

[6] R. Zhao, B. Du, and L. Zhang, "A robust nonlinear hyperspectral anomaly detection approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1227–1234, Apr. 2014.

[7] R. Zhao, B. Du, L. Zhang, and L. Zhang, "Beyond background feature extraction: An anomaly detection algorithm inspired by slowly varying signal analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1757–1774, Mar. 2016.

[8] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[9] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022.

[10] C.-I. Chang and S.-S. Chiang, "Anomaly detection and classification for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 6, pp. 1314–1325, Jun. 2002.

[11] X. Tao, M. E. Paoletti, Z. Wu, J. M. Haut, P. Ren, and A. Plaza, "An abundance-guided attention network for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505414.

[12] R. Zhao, B. Du, L. Zhang, and L. Zhang, "A robust background regression based score estimation algorithm for hyperspectral anomaly detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 122, pp. 126–144, Dec. 2016.

[13] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.

[14] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Aug. 1990.

[15] H. Kwon, S. Z. Der, and N. M. Nasrabadi, "Adaptive anomaly detection using subspace separation for hyperspectral imagery," *Opt. Eng.*, vol. 42, no. 11, pp. 3342–3351, 2003.

[16] D. W. J. Stein, S. G. Beaven, L. E. Hoff, E. M. Winter, A. P. Schaum, and A. D. Stocker, "Anomaly detection from hyperspectral imagery," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 58–69, Jan. 2002.

[17] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[18] A. Banerjee, P. Burlina, and C. Diehl, "A support vector method for anomaly detection in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2282–2291, Aug. 2006.

[19] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[20] M. Vafadar and H. Ghassemian, "Anomaly detection of hyperspectral imagery using modified collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 577–581, Apr. 2018.

[21] H. Su, Z. Wu, Q. Du, and P. Du, "Hyperspectral anomaly detection using collaborative representation with outlier removal," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 5029–5038, Dec. 2018.

[22] S. Feng, S. Tang, C. Zhao, and Y. Cui, "A hyperspectral anomaly detection method based on low-rank and sparse decomposition with density peak guided collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2021.

[23] Z. Wu et al., "Hyperspectral anomaly detection with relaxed collaborative representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5533417.

[24] Y. Xu, Z. Wu, J. Chanussot, and Z. Wei, "Joint reconstruction and anomaly detection from compressive hyperspectral images using Mahalanobis distance-regularized tensor RPCA," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2919–2930, May 2018.

[25] L. Li, W. Li, Y. Qu, C. Zhao, R. Tao, and Q. Du, "Prior-based tensor approximation for anomaly detection in hyperspectral imagery," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 3, pp. 1037–1050, Mar. 2022.

[26] S. Sun, J. Liu, X. Chen, W. Li, and H. Li, "Hyperspectral anomaly detection with tensor average rank and piecewise smoothness constraints," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8679–8692, 2023, doi: 10.1109/TNNLS.2022.3152252.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.

[27] S. Sun, J. Liu, Z. Zhang, and W. Li, "Hyperspectral anomaly detection based on adaptive low-rank transformed tensor," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9787–9799, 2024, doi: 10.1109/TNNLS.2023.3236641.

[28] W. Shang et al., "Hyperspectral anomaly detection via sparsity of core tensor under gradient domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517816.

[29] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[30] T. Cheng and B. Wang, "Graph and total variation regularized low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 391–406, Jan. 2020.

[31] L. Ren, L. Gao, M. Wang, X. Sun, and J. Chanussot, "HADGSM: A unified nonconvex framework for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5503415.

[32] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[33] Y. Li, K. Jiang, W. Xie, J. Lei, X. Zhang, and Q. Du, "A model-driven deep mixture network for robust hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522916.

[34] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.

[35] W. Li, G. Wu, and Q. Du, "Transferred deep learning for anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 597–601, May 2017.

[36] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.

[37] G. Fan, Y. Ma, X. Mei, F. Fan, J. Huang, and J. Ma, "Hyperspectral anomaly detection with robust graph autoencoders," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5511314.

[38] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[39] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. Plaza, "BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531916.

[40] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. J. Plaza, "PDBSNet: Pixel-shuffle downsampling blind-spot reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511914.

[41] Z. Wu et al., "Background-guided deformable convolutional autoencoder for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531816.

[42] S. Lin, M. Zhang, X. Cheng, L. Shi, P. Gamba, and H. Wang, "Dynamic low-rank and sparse priors constrained deep autoencoders for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–18, 2024.

[43] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[44] W. Xie, B. Liu, Y. Li, J. Lei, and Q. Du, "Autoencoder and adversarial-learning-based semisupervised background estimation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5416–5427, Aug. 2020.

[45] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[46] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[47] T. Jiang, W. Xie, Y. Li, J. Lei, and Q. Du, "Weakly supervised discriminative learning with spectral constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6504–6517, Nov. 2022.

[48] W. Xie, X. Zhang, Y. Li, J. Lei, J. Li, and Q. Du, "Weakly supervised low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 51, no. 8, pp. 3889–3900, Aug. 2021.

[49] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Feb. 12, 2024, doi: 10.1109/TNNLS.2024.3355166.

[50] Z. Wu and B. Wang, "Transformer-based autoencoder framework for nonlinear hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508015.

[51] Z. Wang, D. Ma, G. Yue, B. Li, R. Cong, and Z. Wu, "Self-supervised hyperspectral anomaly detection based on finite spatialwise attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502918.

[52] D. Ma, M. Chen, Y. Yang, B. Li, M. Li, and Y. Gao, "Coarse-to-fine multiview anomaly coupling network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507715.

[53] M. E. Paoletti, S. Moreno-Álvarez, Y. Xue, J. M. Haut, and A. Plaza, "AAtt-CNN: Automatic attention-based convolutional neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5511118.

[54] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[55] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[56] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Deep low-rank prior for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5527017.

[57] L. Wang, Y. Wang, Z. Li, C. Wu, M. Xu, and M. Shao, "Eliminating spatial correlations of anomaly: Corner-visible network for unsupervised hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5529114.

[58] W. Liu et al., "UADNet: A joint unmixing and anomaly detection network based on deep clustering for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511419.

[59] J. Wang, J. Sun, Y. Xia, and Y. Zhang, "Dynamic negative sampling autoencoder for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9829–9841, 2022.

[60] P. Gao, T. Ma, H. Li, Z. Lin, J. Dai, and Y. Qiao, "ConvMAE: Masked convolution meets masked autoencoders," 2022, *arXiv:2205.03892*.

[61] S. Woo et al., "ConvNeXt v2: Co-designing and scaling ConvNets with masked autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 16133–16142.

[62] K. Tian, Y. Jiang, Q. Diao, C. Lin, L. Wang, and Z. Yuan, "Designing BERT for convolutional networks: Sparse and hierarchical masked modeling," 2023, *arXiv:2301.03580*.

[63] C. Choy, J. Gwak, and S. Savarese, "4D spatio-temporal ConvNets: Minkowski convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3075–3084.

[64] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351. Cham, Switzerland: Springer, 2015, pp. 234–241.

[65] Z. Tu et al., "MaxViT: Multi-axis vision transformer," in *Proc. 17th Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 459–479.

[66] H. Wang, X. Chen, B. Ni, Y. Liu, and J. Liu, "Omni aggregation networks for lightweight image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22378–22387.

[67] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[68] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[69] S. Li, K. Zhang, P. Duan, and X. Kang, "Hyperspectral anomaly detection with kernel isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 319–329, Jan. 2020.

[70] P. Xiang, S. Ali, S. K. Jung, and H. Zhou, "Hyperspectral anomaly detection with guided autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5538818.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 10:22:05 UTC from IEEE Xplore. Restrictions apply.